File 71: Creekside Stream Monitoring Field Data Sheet

Field Data Collection Form for Fluvial Geomorphology & Wild Trout Spawning Assessments

> ***Stream Monitoring Field Form Standard**: This document represents the official **Creekside Stream Monitoring Field Data Sheet (File 71)** for the **Anderson Creek Showcase Restoration Project (Roya's Cabin Reach)**. It provides Hunter Morris, Trout Unlimited, and university interns (UGA Warnell, Georgia Tech) with a ready-to-print, structured field data sheet to systematically record Wolman pebble counts, EPA RBP benthic macroinvertebrates, physical water parameters, and trout spawning nests (redds).*

I. Project & Station Header Information

- **Project Name / Reach**: Anderson Creek Showcase Restoration (Roya's Cabin Reach)
- **Station ID (HUC-8)**: GA-03150105 / Station # ________ (e.g. Monument #01)
- **Field Observers / Crew**: ____
- **Sampling Date**: ____ / ____ / 202__ **Time**: ____ : ____ AM/PM
- **Weather Conditions**: [ ] Sunny [ ] Partly Cloudy [ ] Overcast [ ] Raining
- **Air Temperature**: ______ °F **Recent Precipitation (24 hr)**: ______ inches

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II. Hydrological & Biochemical Water Quality Parameters Log

Equipment Calibrated Prior to Field Work? [ ] Yes [ ] No (YSI ProDSS Calibration Complete)

Parameters	Unit	Reading #1	Reading #2	Reading #3	Arithmetic Mean	Standard Deviation
Water Temperature	°F (°C)					
Dissolved Oxygen	mg/L					
Turbidity Index	NTU					
Specific pH	s.u.					

Parameters	Unit	Reading #1	Reading #2	Reading #3	Arithmetic Mean	Standard Deviation
Stream Flow Velocity	ft/s (m/s)					
Gauge Stage Height	ft (m)					

III. Wolman Pebble Count Tally Grid

Measure 100 random bed substrate particles across the cross-section riffle profile to compute median diameter (D■■)

Particle Class	Size Range (mm)	Sieve Classification	Tally (e.g. III, IIII, etc.)	Total Count
Silt / Clay	< 0.062 mm	Silt		
Very Fine Sand	0.062 - 0.125 mm	Sand		
Fine Sand	0.125 - 0.25 mm	Sand		
Medium Sand	0.25 - 0.50 mm	Sand		
Coarse Sand	0.50 - 1.0 mm	Sand		
Very Coarse Sand	1.0 - 2.0 mm	Sand		
Very Fine Gravel	2.0 - 4.0 mm	Pebble (Spawning)		
Fine Gravel	4.0 - 5.7 mm	Pebble (Spawning)		
Medium Gravel	5.7 - 8.0 mm	Pebble (Spawning)		
Medium Gravel	8.0 - 11.3 mm	Pebble (Spawning)		
Coarse Gravel	11.3 - 16.0 mm	Pebble (Spawning)		
Coarse Gravel	16.0 - 22.6 mm	Pebble (Spawning)		
Very Coarse Gravel	22.6 - 32.0 mm	Pebble (Spawning)		
Very Coarse Gravel	32.0 - 45.0 mm	Pebble (Spawning)		
Very Coarse Gravel	45.0 - 64.0 mm	Pebble (Spawning)		
Small Cobble	64 - 90 mm	Cobble		
Medium Cobble	90 - 128 mm	Cobble		

Particle Class	Size Range (mm)	Sieve Classification	Tally (e.g. III, IIII, etc.)	Total Count
Large Cobble	128 - 180 mm	Cobble		
Very Large Cobble	180 - 256 mm	Cobble		
Small Boulder	256 - 512 mm	Boulder		
Medium Boulder	512 - 1024 mm	Boulder		
Large Bedrock	> 1024 mm	Bedrock		
TOTAL COUNT	—	—	—	**100**

IV. EPA RBP Benthic Macroinvertebrate Tally Card

Conduct a standard 3-kick net composite sample in gravel riffles. Sort and count sensitive EPT aquatic insects

1. Group 1: Sensitive Taxa (EPT Index - Indicator of clean, cold trout waters)

- **Mayflies (Ephemeroptera)**:
- *Heptageniidae* (Flathead Mayfly): ______ *Baetidae* (Minnow Mayfly): ______ *Ephemerellidae* (Spiny Crawler): ______
- **Stoneflies (Plecoptera)**:
- *Perlidae* (Common Stonefly): ______ *Peltoperlidae* (Roach-like Stonefly): ______ *Nemouridae* (Forest Stonefly): ______
- **Caddisflies (Trichoptera)**:
- *Hydropsychidae* (Net-spinner): ______ *Glossosomatidae* (Saddlecase): ______ *Limnephilidae* (Northern Case): ______

2. Group 2: Moderately Tolerant Taxa

- **Dragonflies / Damselflies (Odonata)**: ______
- **Crayfish (Decapoda)**: ______
- **Dobsonflies / Hellgrammites (Corydalidae)**: ______
- **Blackflies (Simuliidae)**: ______

3. Group 3: Highly Tolerant Taxa (Indicator of siltation and hypoxia)

- **Midges (Chironomidae)**: ______
- **Aquatic Worms (Oligochaeta)**: ______
- **Leeches (Hirudinea)**: ______

4. Biological Health Calculations

- **Total Taxa Richness**: ______ distinct families observed.

- **EPT Index Score**: _______ (Sum of Mayfly + Stonefly + Caddisfly families).
- *Evaluation*: EPT >= 12 = Excellent; 8 - 11 = Good; 4 - 7 = Fair; <= 3 = Poor.

V. Wild Trout Spawning Nest (Redd) Log Sheet

Walk the riffle meanders in late autumn. Identify and map clean gravel pockets used by spawning trout

Redd #	Geospatial Coordinates (Lat/Lon)	Water Depth (ft)	Flow Velocity over Redd (ft/s)	Redd Length (ft)	Redd Width (ft)	Parent Trout Species (Brown/Brook)
01						
02						
03						
04						
05						
06						
07						
08						

Redd Mapping Field Notes

(Describe any sediment deposits near redds, presence of spawning parent trout, canopy shading levels, or wood debris structures nearby)

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Prepared by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin.